

Task 1:- ER Diagram

Attributes of each entity:-

①. Student

- student_id (PRIMARY KEY)
- name
- email (UNIQUE)
- age
- gender
- dept_id (FOREIGN KEY → Department.dept_id)

②. Department

- dept_id (PRIMARY KEY)
- dept_name
- building

③. Teacher

- teacher_id (PRIMARY KEY)
- name
- email (UNIQUE)
- designation
- dept_id (FOREIGN KEY → Department.dept_id)

④. Course

- course_id (PRIMARY KEY)
- course_name
- credits
- dept_id (FOREIGN KEY → Department.dept_id)
- teacher_id (FOREIGN KEY → Teacher.teacher_id)

⑤. Enrollment (Associative Entity)

- student_id (PRIMARY KEY, FOREIGN KEY → student.student_id)
- course_id (PRIMARY KEY, FOREIGN KEY → course.course_id)
- semester
- grade

⑥. Exam

- exam_id (PRIMARY KEY)
- exam_type
- course_id (FOREIGN KEY → course.course_id)

⑦. Result (Associative entity)

- student_id (PRIMARY KEY, FOREIGN KEY → student.student_id)
- exam_id (PRIMARY KEY, FOREIGN KEY → Exam.exam_id)
- marks

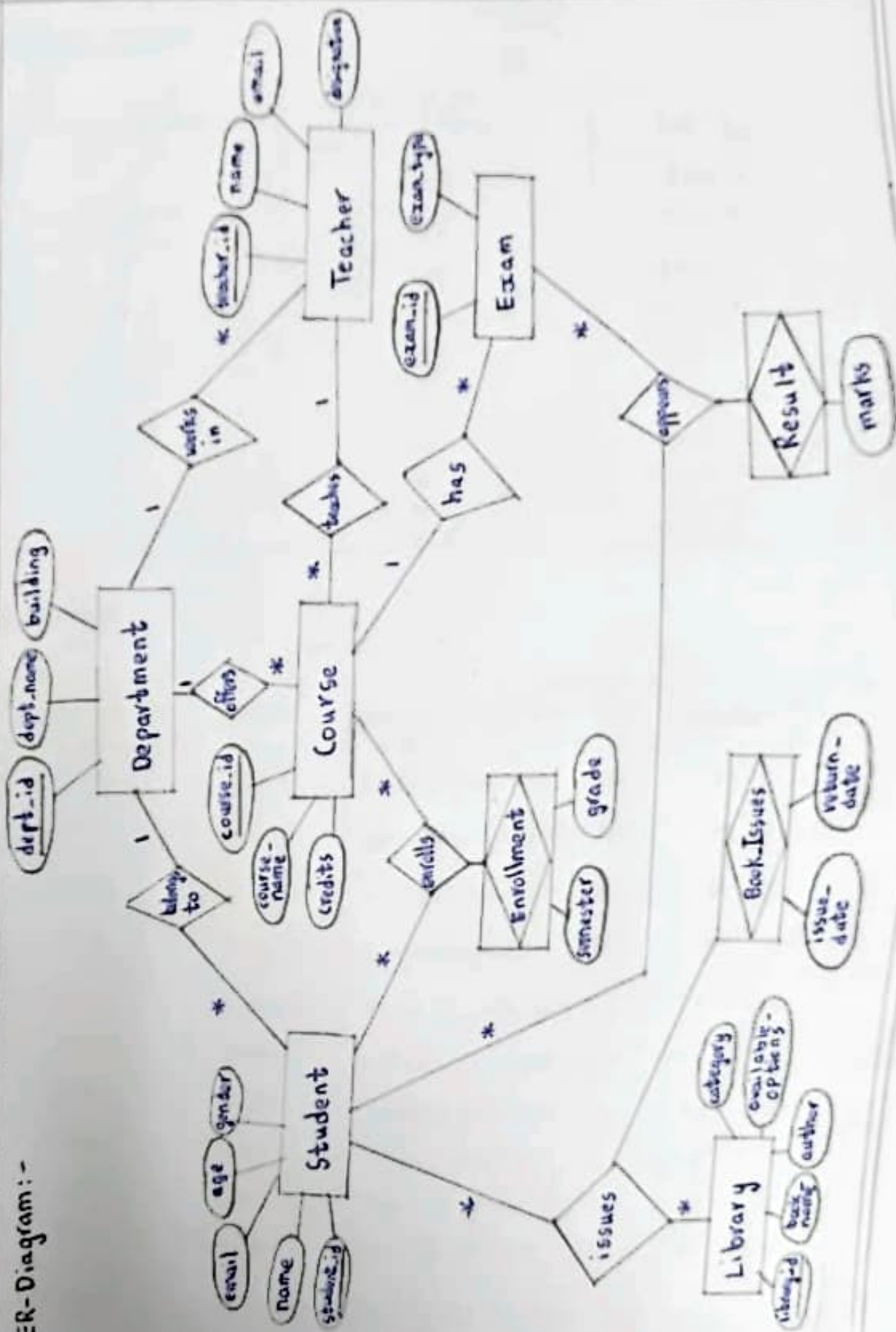
⑧. Library

- library_id (PRIMARY KEY)
- book_name
- author
- category
- available_copies

⑨. Book-Issue

- issue_id (PRIMARY KEY)
- student_id (FOREIGN KEY → student.student_id)
- library_id (FOREIGN KEY → Library.library_id)
- issue_date
- return_date

ER-Diagram :-



Task 2:- Relations/Relationships into tables.

①. Department

dept_id (PRIMARY KEY)	dept_name	building
1	'CSE'	'Block A'
2	'ECE'	'Block B'
3	'EEE'	'Block C'
4	'ME'	'Block D'
5	'CE'	'Block E'
6	'IT'	'Block F'
7	'AI'	'Block G'
8	'DS'	'Block H'
9	'Bio'	'Block I'
10	'Chem'	'Block J'

②. Student

student_id (PK)	name	email (UNIQUE)	age	gender	dept_id (FK)
101	'Rohit'	'rohit@gmail.com'	21	'M'	1
102	'Aman'	'aman@gmail.com'	22	'M'	2
103	'Sneha'	'sneha@gmail.com'	20	'F'	1
104	'Priya'	'priya@gmail.com'	21	'F'	3
105	'Karan'	'karan@gmail.com'	23	'M'	4
106	'Neha'	'neha@gmail.com'	20	'F'	5
107	'Arjun'	'arjun@gmail.com'	22	'M'	6
108	'Pooja'	'pooja@gmail.com'	21	'F'	7
109	'Rahul'	'rahul@gmail.com'	23	'M'	8
110	'Simran'	'simran@gmail.com'	20	'F'	9

③. Teacher

teacher_id (PK)	name	email (UNIQUE)	designation	dept_id (FK)
201	'Dr. Sharma'	'sharma@gmail.com'	'Professor'	1
202	'Dr. Verma'	'verma@gmail.com'	'Associate Prof'	2
203	'Dr. Singh'	'singh@gmail.com'	'Professor'	3
204	'Dr. Mehta'	'mehta@gmail.com'	'Assistant Prof'	4
205	'Dr. Rao'	'rao@gmail.com'	'Professor'	5
206	'Dr. Iyer'	'iyer@gmail.com'	'Assistant Prof'	6
207	'Dr. Khan'	'khan@gmail.com'	'Professor'	7
208	'Dr. Das'	'das@gmail.com'	'Associate Prof'	8
209	'Dr. Roy'	'roy@gmail.com'	'Professor'	9
210	'Dr. Jain'	'jain@gmail.com'	'Assistant Prof'	10

④. Courses

course_id (PK)	course_name	credits	dept_id (FK)	teacher_id (FK)
301	'DBMS'	4	1	201
302	'Signals'	3	2	202
303	'Machines'	4	3	203
304	'Thermodynamics'	3	4	204
305	'Structures'	4	5	205
306	'Networks'	3	6	206
307	'AI basics'	4	7	207
308	'Data Science'	4	8	208
309	'Biology 101'	3	9	209
310	'Chemistry 101'	3	10	210

⑤. Enrollment

student_id (PK)	course_id (PK)	Semester	grade
101	301	1	'A'
102	302	1	'B'
103	301	2	'A'
104	303	1	'C'
105	304	2	'B'
106	305	1	'A'
107	306	2	'B'
108	307	1	'A'
109	308	2	'C'
110	309	1	'B'

⑥. Exam

exam_id (PK)	exam_type	course_id (FK)
401	'Mid'	301
402	'End'	302
403	'Mid'	303
404	'End'	304
405	'Mid'	305
406	'End'	306
407	'Mid'	307
408	'End'	308
409	'Mid'	309
410	'End'	310

⑦ Result

student_id (PK)	exam_id (PK)	marks
101	401	85
102	402	78
103	401	90
104	403	65
105	404	70
106	405	88
107	406	76
108	407	92
109	408	60
110	409	75

⑧. Library

library_id (PK)	book_name	author	category	available
501	'DBMS Book'	'Korth'	'CS'	5
502	'Signals Book'	'Oppenheim'	'ECE'	4
503	'Machines Book'	'Bimbhra'	'EEE'	6
504	'Thermo Book'	'Cengel'	'ME'	3
505	'Structures Book'	'Hibbeler'	'CE'	5
506	'Networks Book'	'Tanenbaum'	'IT'	4
507	'AI Book'	'AI Russell'	'AI'	6
508	'DS Book'	'DS James'	'DS'	5
509	'Bio Book'	'Bio Campbell'	'Bio'	3
510	'Chem Book'	'Chem Morrison'	'Chem'	4

⑨. Book - Issue

issue_id (Pk)	student_id (FK)	library_id (FK)	issue_date	return_date
601	101	501	'2026-03-01'	'2026-03-10'
602	102	502	'2026-03-02'	'2026-03-12'
603	103	503	'2026-03-03'	'2026-03-13'
604	104	504	'2026-03-04'	'2026-03-14'
605	105	505	'2026-03-05'	'2026-03-15'
606	106	506	'2026-03-06'	'2026-03-16'
607	107	507	'2026-03-07'	'2026-03-17'
608	108	508	'2026-03-08'	'2026-03-18'
609	109	509	'2026-03-09'	'2026-03-19'
610	110	510	'2026-03-10'	'2026-03-20'